

RISHABH PATIDAR

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EDUCATION

Madhav Institute of Technology and Science

Gwalior, MP

Bachelor of Technology in Information Technology | **CGPA: 8.124**

Expected: May 2027

Relevant Coursework: Machine Learning, Data Structures & Algorithms, DBMS, Artificial Intelligence, Statistics

TECHNICAL SKILLS

Languages: Python, SQL, Java, C/C++

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy, Statsmodels, Excel

ML / Modelling: Scikit-learn, XGBoost, Random Forest, Gradient Boosting, KMeans, PCA, SHAP, SMOTE, RFM Analysis, Feature Engineering, A/B Testing, Hypothesis Testing

Deployment & Tools: Streamlit, FastAPI, Groq API (LLaMA-3.3-70B), Git, Jupyter, Pydantic V2, Google Colab

Computer Vision (Supporting): InsightFace, MediaPipe, OpenCV — applied in forensic analytics pipeline

EXPERIENCE

Undergraduate Research Assistant

Jan – Apr 2026

Madhav Institute of Technology and Science

Gwalior, MP

- Conducted end-to-end exploratory data analysis on event marketing datasets covering 50+ features, identifying key footfall drivers through correlation analysis and domain-informed feature selection, reducing model noise by 30%
- Built a two-stage XGBoost Regressor pipeline ($R^2 = 0.96$) with a Dynamic Physical Clamping algorithm enforcing venue-capacity constraints at inference, achieving zero invalid output predictions across all test cases
- Integrated LLaMA-3.3-70B via Groq API to automate brand-event synergy scoring and generate structured JSON negotiation reports, deployed as a stateless FastAPI microservice enabling data-driven sponsorship ROI estimation

PROJECTS

Customer Segmentation & Churn Prediction | *Scikit-learn, XGBoost, SHAP, SMOTE, Streamlit, SQL*

- Engineered 22 RFM and behavioural features from 2.5M+ raw transactions by designing a modular SQL-to-Pandas ETL pipeline, reducing feature computation time by 40% and enabling scalable segment re-generation
- Applied KMeans ($k=4$) clustering to surface 4 actionable customer segments including a 15.7% churn-rate cohort, then trained a Random Forest classifier (ROC-AUC 0.8793) with SMOTE cross-validation to handle class imbalance
- Used SHAP TreeExplainer to confirm recency as the #1 churn driver, translating probability scores into \$233,917 revenue-at-risk quantification across 392 high-risk households and presenting findings via an interactive Streamlit dashboard

CinemaIQ — Box Office Intelligence Platform | *Python, XGBoost, Scikit-learn, Streamlit, Plotly, Groq API*

- Designed novel actor, director, and composer power indices from historical performance data using weighted aggregation and multi-hot genre encoding, creating 30+ domain-specific features for South Asian film revenue prediction
- Benchmarked 5 regression models (Linear, Ridge, Random Forest, Gradient Boosting, XGBoost) with consistent train/val/test splits and log1p target transformation, selecting XGBoost based on lowest RMSE and best holdout generalisation
- Deployed a 3-tab Streamlit dashboard integrating Pessimistic/Base/Optimistic scenario forecasting, ROI gauges, and LLaMA 3.3-powered AI commentary via Groq API, enabling non-technical stakeholders to interpret predictions interactively

Video Target Identification System | *Python, InsightFace, MediaPipe, OpenCV, Streamlit*

- Built a biometric analytics pipeline fusing InsightFace 512-d face embeddings and MediaPipe 12-d pose/gait descriptors via weighted cosine similarity (face 0.7, pose 0.3) to enable person re-identification across multiple CCTV feeds
- Engineered consecutive-frame validation logic and configurable confidence thresholds to suppress false positives, improving identification precision by reducing spurious matches in low-quality footage scenarios
- Deployed a Streamlit forensic dashboard supporting multi-video batch scanning, structured evidence export (PDF report, CSV log, ZIP of face crops), and visual similarity score analytics for downstream investigative review

ACHIEVEMENTS

1st Runner-Up — SSH '26 National Hackathon | SSH | Feb 2026

Top Performer in WebDev Track — Hacksagon | ABV IIITM Gwalior | Apr 2026

Finalist — ABV-IIITM Hackatron | ABV-IIITM (Powered by GitHub) | Oct 2025